

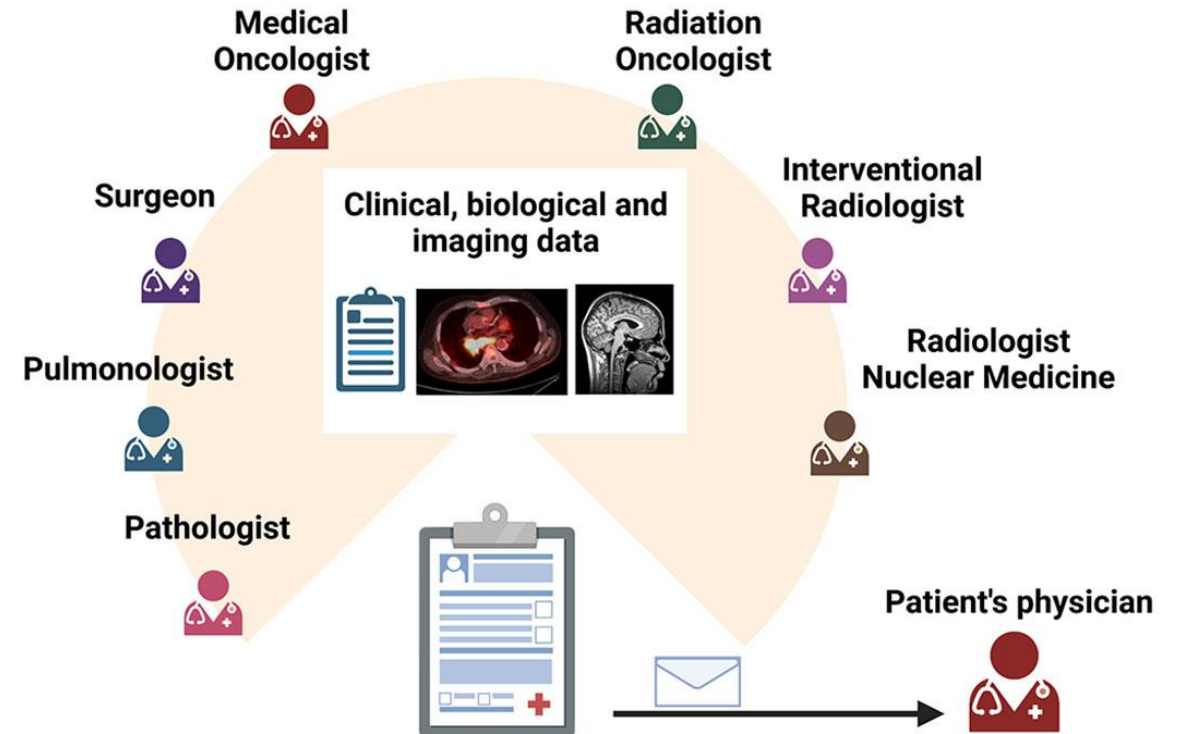
The Virtual Tumor Board: Two Specialized Agents, One Treatment Plan

A multi-agent clinical reasoning simulation using Qwen and DeepSeek

Overview

Tumor Boards

- **Cancer treatment decisions are multi-perspective.**
- A tumor board combines different specialist views:
 - Oncologist: treatment strategy, prognosis, toxicity, guidelines
 - Pathologist: tissue diagnosis, grade, margins, biomarkers
 - Shared goal: reach a coherent treatment recommendation
- **Key idea:**
 - *A single generic LLM may give a plausible answer, but it may not reproduce specialist disagreement or structured revision.*

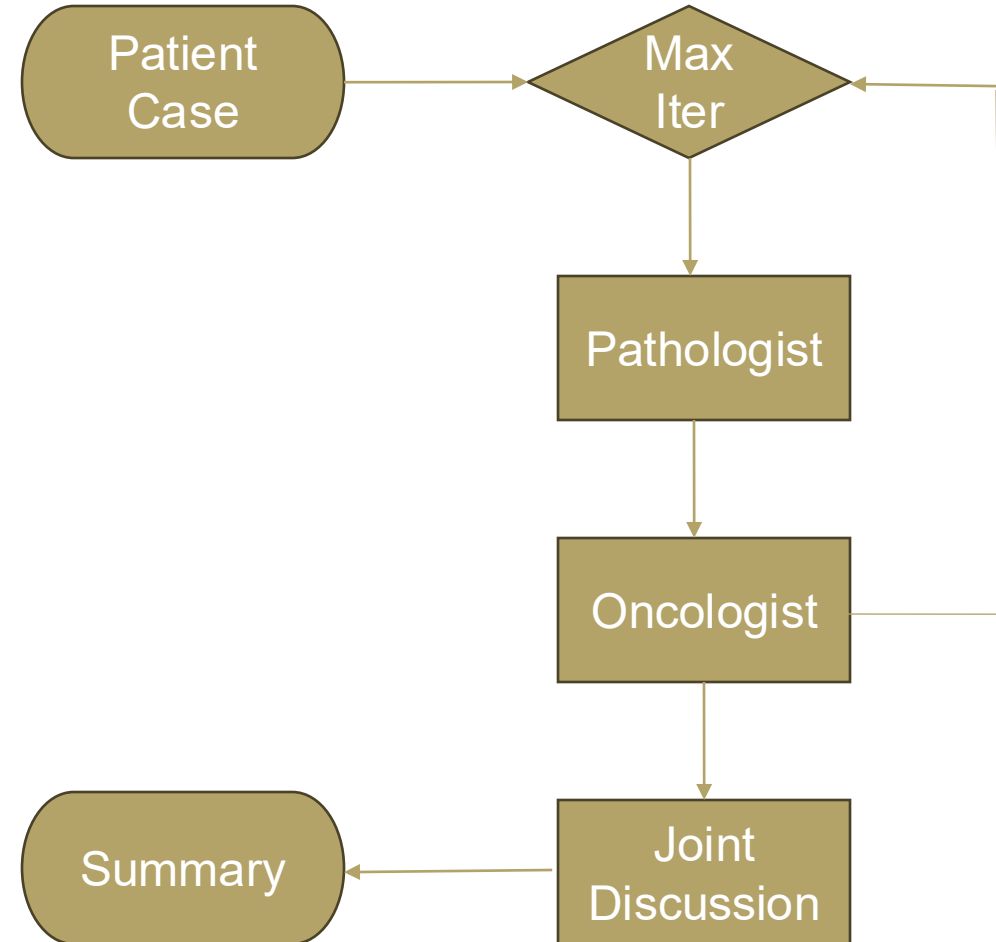


Assignment Objective

- Goal:
 - Build a two-agent simulated tumor board.
- System requirements:
 - Oncologist Agent runs on Qwen
 - Pathologist Agent runs on DeepSeek
 - Agents discuss realistic fictional cancer cases
 - The discussion follows a structured protocol
 - Each case ends with a joint treatment summary
- My focus:
 - Can role-specialized LLM agents produce useful clarification, disagreement, and treatment revision?

System Overview

- System flow:
 - Input a fictional patient case
 - Pathologist interprets pathology findings
 - Oncologist proposes treatment strategy
 - Agents exchange clarification and caveats
 - System generates a joint final summary
- Inputs:
 - Clinical summary
 - Imaging findings
 - Pathology report
- Outputs:
 - Diagnosis
 - Treatment recommendation
 - Key uncertainties
 - Follow-up tests



Agent Role Separation

Oncologist Agent — Qwen	Pathologist Agent — DeepSeek
Clinical stage	Histology
Prognosis	Tumor grade
Treatment options	Margins
Toxicity	Receptor status
Quality of life	Molecular biomarkers
Guidelines	Tissue adequacy

- The agents are not two copies of the same chatbot.
- Each agent is designed to contribute a different type of reasoning.

Prompt Design

Oncologist Agent – Qwen

- Role boundary:
 - Does not diagnose from tissue
 - Focuses on treatment strategy, staging, prognosis, toxicity, and patient fitness
- Reasoning:
 - 1. Determine clinical stage or likely stage
 - 2. Determine treatment intent
 - 3. Identify pathology-driven treatment factors
 - 4. Compare reasonable treatment options
 - 5. Weigh benefit against toxicity and quality of life
 - 6. State current recommendation
- Behavior rules:
 - Use the Pathologist's confirmed facts
 - Do not invent pathology details
 - Ask at least one treatment-relevant pathology clarification
 - State uncertainty before finalizing treatment

Pathologist Agent – DeepSeek

- Role boundary:
 - Does not choose the full systemic therapy plan
 - Focuses on tissue findings, histology, grade, margins, biomarkers, and sample adequacy
- Reasoning:
 - Separate confirmed findings, inferred implications, and missing or pending information
 - Stay grounded in what the specimen directly shows
 - Do not infer missing biomarkers or molecular results
- Behavior rules:
 - Flag insufficient tissue, missing stains, pending biomarkers, or ambiguous morphology
 - Challenge treatment assumptions when tissue evidence is incomplete
 - Explain treatment implications only through pathology findings

Main Takeaway: Prompt design prevents the two agents from collapsing into the same generic medical chatbot.

Conversation Protocol

The agents do not chat freely.

The discussion follows a five-round protocol.

- Pathologist opens
 - Interprets tumor type, grade, biomarkers, and ambiguities.
- Oncologist responds
 - Gives initial treatment hypothesis and asks a pathology question.
- Pathologist clarifies
 - Answers the question and flags complicating findings.
- Oncologist revises
 - Updates the treatment recommendation.
- Joint summary
 - Both perspectives contribute to final diagnosis, plan, uncertainties, and tests.

Case Walkthrough

Test Cases

Case	Cancer Type	Purpose
Case 1	Early-stage breast cancer	Straightforward receptor-driven treatment
Case 2	Lung adenocarcinoma	Missing biomarkers and diagnostic uncertainty
Case 3	Colorectal cancer	Pathology changes treatment direction

Case 3: Stage II Colon Cancer Where MMR Status Changes Adjuvant Thinking

Clinical summary

- 46-year-old woman
- Iron-deficiency anemia and right-sided abdominal discomfort
- Father had colorectal cancer at age 49
- ECOG 0

Imaging

- Localized ascending colon mass
- Small regional nodes
- No distant metastases
- CT chest negative

Pathology

- Right hemicolectomy specimen
- Moderately differentiated adenocarcinoma, 4.8 cm
- Invades through muscularis propria into pericolonic tissue
- Margins negative
- 0/24 lymph nodes positive
- No LVI or PNI
- Low tumor budding
- Loss of MLH1 and PMS2; intact MSH2/MSH6
- BRAF V600E and MLH1 promoter methylation pending

This case tests whether the agents recognize that dMMR status changes the adjuvant treatment decision. (No Chemotherapy)

Round 1: Pathologist Identifies the Treatment-Relevant Pathology (Pathologist interpretation)

Confirmed findings

- Stage II colon adenocarcinoma: pT3 N0
- Margins negative
- 0/24 lymph nodes positive
- No lymphovascular or perineural invasion
- Low tumor budding

Key biomarker

- Loss of MLH1 and PMS2
- Intact MSH2 and MSH6
- Interpreted as dMMR profile

Uncertainties

- BRAF V600E pending
- MLH1 promoter methylation pending
- Lynch syndrome cannot be confirmed or excluded yet
- Tumor content for molecular testing not explicitly reported

The Pathologist reframes the case from generic stage II colon cancer to stage II dMMR colon cancer.

"dMMR status in stage II colon cancer is a strong favorable prognostic factor"

Round 2: Oncologist Converts Pathology Into Treatment Strategy (Oncologist reasoning)

Stage and intent

- pT3 N0, Stage IIA
- No metastatic disease
- Curative-intent setting

Treatment direction

- Observation favored
- No adjuvant fluoropyrimidine chemotherapy
- Avoid unnecessary toxicity in a young ECOG 0 patient

Why pathology matters

- dMMR changes adjuvant chemotherapy thinking
- MLH1/PMS2 loss raises hereditary risk questions
- BRAF and MLH1 methylation results guide sporadic vs Lynch (genetic) pathway

Clarifying question

- Was the tumor tissue for BRAF and MLH1 methylation testing enriched enough for reliable molecular results?

The Oncologist uses the pathology facts to move from stage-based treatment to biomarker-guided management.

"Recommend observation (no adjuvant chemotherapy) — provisionally, pending resolution of the pending molecular studies."

Round 3: Pathologist Flags a Preanalytic Caveat (Pathologist clarification)

Answered staging concern

- Wording is strongly consistent with pT3
- Serosal involvement is not explicitly stated
- Addendum could remove residual ambiguity

Answered molecular testing concern

- Tumor cellularity for BRAF / MLH1 methylation testing is not documented
- Macrodissection or microdissection status is not reported
- Low tumor content could make molecular results unreliable

Additional uncertainty

- MSI-PCR not reported
- IHC is usually sufficient, but rare discordance is possible

The molecular results may not be clinically interpretable unless testing adequacy is verified.

"The pathology report provided does not state whether the sample used for BRAF and MLH1 methylation testing was macro- or micro-dissected, nor does it give a tumor cellularity estimate."

Round 4: Oncologist Updates the Management Pathway (What changed after the Pathologist's clarification?)

Before Round 3

- Observation favored
- Await BRAF and MLH1 methylation results
- Use results to decide sporadic vs Lynch pathway

After Round 3

- Observation remains the treatment recommendation
- But the plan is not fully actionable until molecular testing adequacy is confirmed
- BRAF and MLH1 methylation results must be interpreted only if tissue adequacy is verified
- If inadequate, repeat testing on a better block may be needed

The treatment recommendation did not switch to a different drug.

The reasoning changed: from “wait for results” to “verify whether the results will be reliable.”

“Observation remains the definitive recommendation—but implementation is paused until analytic validity of molecular testing is confirmed.”

Final Joint Summary: Biomarker-Guided Observation

Working diagnosis

- Stage IIA right-sided colon adenocarcinoma
- pT3 N0 M0
- dMMR by MLH1/PMS2 loss

Treatment recommendation

- Observation only
- No adjuvant chemotherapy indicated

Key follow-up actions

- Confirm molecular testing adequacy
- Await BRAF V600E and MLH1 promoter methylation results
- Request pathology addendum for serosal status if needed
- Refer to genetics if molecular pattern supports possible Lynch syndrome

Why the discussion converged

- dMMR became the dominant treatment-relevant pathology finding
- The final plan combined oncology management with pathology-driven diagnostic stewardship

Final plan: no chemotherapy, active surveillance, and molecular/genetic clarification.

“Observation only — no adjuvant chemotherapy indicated.”

Does Model Choice Matter?

Observation	Qwen (Oncologist)	DeepSeek (Pathologist)
Reasoning focus	Treatment strategy	Biomarkers
Uncertainty handling	Converts uncertainty into next clinical steps	Flags missing or unreliable pathology evidence
Verbosity	Long treatment rationale, can argue on the same concern	Long pathology caveat
Clinical grounding	Management-oriented	Evidence-oriented
Output-style	Emojis with deeper hierarchy	Bullet point and simple list
Contribution	Revises treatment pathway	Challenges evidence quality